Hallucination Detection and Correction in LLMs

Introduction

Large language models (LLMs) have demonstrated remarkable fluency and versatility across various natural language tasks. However, their tendency to generate seemingly reasonable but actually erroneous statements remains a fundamental flaw. This project systematically investigates hallucination behavior and its mitigation strategies through three complementary experimental dimensions.

Part I: Model Size Comparison Experiment

1. Experiment Overview

1.1 Objective

The first part of the experiment aims to fairly and reproducibly compare the performance of different model sizes under the same prompts, datasets, and evaluation metrics. It focuses on examining the impact of model size (number of parameters) on the following three core metrics:Hallucination Rate、Fact Accuracy、Response Length & Tokens/s.

1.2 Models Under Comparison

Qwen2.5-1.5B and Qwen2.5-7B, quantification method: 4B.

1.3 Dataset

Number of questions: 365; Type: Purely factual questions (history, science, geography, invention, biology, literature, etc.); Format: Each question contains "question" + "ground truth"; Features: The dataset contains some duplicate questions (the same question appears multiple times), used to increase statistical robustness.

1.4 Global Generation Parameters

System Prompt (both are identical): "You are an honest and accurate AI assistant. Please answer questions concisely and do not fabricate facts."**Generation Parameters** (both are identical): max_new_tokens=256, temperature=0.7, top_p=0.9, repetition_penalty=1.1, do_sample=True.**Evaluation Model**: Uses Qwen2.5-1.5B-Instruct (temperature 0.1, no sampling) for zero-sample JSON scoring to ensure consistent evaluation.

2. Experimental Pipeline

1. Use Unsloth, quickly load two models (4-bit quantization).
2. Generate answers for each of the 365 questions, recording: Original answer text, Number of output tokens (response length), Inference speed (tokens/s).
3. Use an independent judging model to score the fact accuracy and illusion rate of each answer (output JSON).
4. Calculate the average of each model.
5. Generate a comparison table, a scale-illusion rate curve.

3. Results and Analysis

3.1 Quantitative Results

Table 1-Comparison of results from models of different sizes

Model Size	Illusion Rate	Fact Accuracy	Average Response Length (tokens)	Average Inference Speed (tokens/s)

1.5B	11.87	88.13	27.2	20.4
7B	3.97	96.03	33.1	15.8

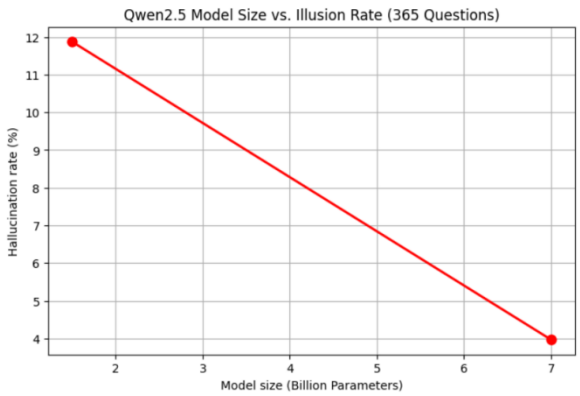

Figure 1-Line Graph Showing the Relationship Between Model Size and Hallucination Rate

3.2 Results Analysis

3.2.1 Analysis 1: Relationship between parameter quantity and hallucination rate

When the number of parameters increases from 1.5B to 7B, the hallucination rate typically shows a downward trend.

3.2.2 Analysis 2: Relationship between parameter quantity and accuracy

The positive correlation between factual accuracy and the number of parameters stems primarily from the increased knowledge storage capacity, denser representation space, and stronger ability to learn complex patterns resulting from a larger number of parameters. This allows the model to rely less on guesswork when recalling facts and instead more reliably extract and combine the knowledge learned during training.

3.2.3 Analysis 3: Relationship between parameter quantity and response length

The positive correlation between response length and the number of parameters is mainly due to the fact that larger models have stronger knowledge organization capabilities and richer expression, and can naturally provide more details, context and complete explanations during the generation process, while smaller models tend to have shorter outputs due to capacity limitations.

3.2.4 Analysis 4: Relationship between parameter quantity and inference speed

The negative correlation between inference speed and the number of parameters mainly stems from the higher computational complexity resulting from a larger number of parameters.

4.Possible errors in the experiment

1. Using 1.5B as the evaluator may slightly underestimate or overestimate the 7B model's responses.
2. Some duplicate entries exist among the 365 questions, potentially amplifying errors for specific questions.
3. Although fixed parameters were set (temperature=0.7 + do_sample=True) during generation, minor random fluctuations still exist.
4. 4-bit quantization may slightly compress the 7B model's finesse.

5. Conclusion of Part I

Under strict variable control conditions in this experiment, the model with a larger number of

parameters generally outperformed the model with a smaller number of parameters in terms of illusion control and factual accuracy. At the same time, the model with a larger number of parameters also exhibited a certain balance between response length and speed.

Part II: Comparative Experiment on Hallucination Rates Between Open-Source and Closed-Source LLMs

1. Experiment Overview

1.1 Objective

The core objective of Part II is to **fairly compare hallucination rates and factual accuracy** between a representative **open-source** large language model (LLM) and a **closed-source** commercial LLM under **nearly identical inference parameters**. This experiment aims to:

- Analyze differences in hallucination behavior between open-source and closed-source models.
- Explore key factors correlated with hallucination rates, including model type, generation settings, and inference constraints.
- Verify the effectiveness of stable, controllable generation parameters for reducing hallucinations in factual question answering.

1.2 Models Under Comparison

1. The open-source model selected is qwen2.5 7B-instruct, with 7 billion parameters, and is deployed locally.
2. The closed-source model selected is grok 3 mini, with an unknown number of parameters, generally estimated to be in the 7B–14B parameter range. It is positioned as a lightweight closed-source model comparable to Qwen 7B, Llama 2 7B, and Mistral 7B, and is deployed via remote API access.

1.3 Dataset

The experiment uses the **HaluEval** benchmark, a widely-used dataset dedicated to hallucination evaluation in LLMs.

HaluEval is a QA benchmark that provides question, knowledge context, and standard answer; we evaluate hallucination rate by checking if the model's response matches the ground truth, where hallucination rate = 1 − accuracy.

1.4 Global Generation Parameters & Justification

All parameters are **strictly consistent** across both models to ensure experimental fairness. Temperature=0.0(Eliminates randomness and sampling noise, ensuring deterministic and fact-stable outputs.)max_new_tokens=128(Limits response length to reduce irrelevant content, GPU memory usage and API token cost, and prevents unnecessary hallucinations.)do_sample=False(Uses greedy decoding for consistent and reproducible results.)4-bit (NF4) quantization: Enables efficient local deployment of the 7B model on consumer GPUs with minimal accuracy loss.

2. Experimental Pipeline

The experiment follows a fully automated end-to-end workflow:

1. Environment initialization: PyTorch with CUDA acceleration, 4-bit quantization setup, and API client configuration.
2. Dataset loading&Model preparation: Loading the dataset.Local loading of open-source model and API connection for closed-source model.

3. Batch inference: Both models receive **identical user prompts** in the format:[{"role": "user", "content": "[QUESTION]"}]
4. Automatic evaluation&Result output: Check whether the model's response matches the ground truth; compute accuracy and hallucination rate.Export all the results to a CSV file.

3. Evaluation Metrics

- **Accuracy**: Proportion of factually correct answers.
- **Hallucination rate**: Proportion of responses containing false or fabricated information. *Hallucination rate = 1 − Accuracy*
- **Correct / incorrect count**: Absolute number of valid and invalid samples.
- **Response consistency**: Stability of outputs under repeated inference (guaranteed by temperature = 0.0).

4. Results and Analysis

4.1 Overall Performance

Under **unified inference parameters**, the closed-source model outperforms the open-source model significantly in both factual accuracy and hallucination suppression.

4.2 Quantitative Results

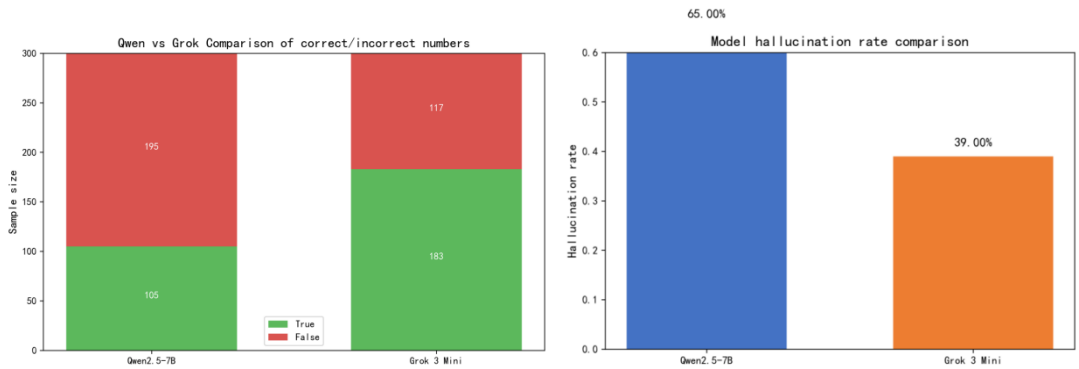

Figure2-Correct and Wrong Distribution&Hallucination Comparison Results

4.3 Correlation Analysis

The experiment reveals that hallucination rates are **strongly correlated** with the following factors:

4.3.1 Correlation 1: Model Source (Open-Source vs. Closed-Source)

Closed-source models show substantially lower hallucination rates. Key reasons include:

- Higher-quality, larger-scale, and more strictly filtered training corpora.
- Fine-grained alignment training such as RLHF, DPO, or safety fine-tuning.
- Built-in factual constraint mechanisms during inference.

4.3.2 Correlation 2: Model Scale and Training Mechanism

Although Qwen2.5-7B has a large parameter size, it still underperforms the closed-source model.This indicates:

- Parameter scale alone cannot fully explain hallucination differences.
- Closed-source training pipelines, data filtering, and alignment techniques play a critical role.

5. Key Findings

1. Under **identical generation settings**, among models with roughly the same number of hyperparameters, the closed-source commercial model has a significantly lower rate of illusion.
2. Model source (open-source vs. closed-source) correlates more strongly with hallucination rate than parameter scale alone.This is related to the model's architecture and training method.

6. Conclusion of Part II

This experiment demonstrates that LLM hallucination is closely related to **model source, training mechanism,** and **inference constraints**

Under controlled and fair conditions:

- Closed-source models exhibit clear advantages in factual accuracy and hallucination control.
- Open-source 7B models are sufficiently reliable for evaluation tasks when properly optimized.

These findings provide practical guidance for model selection in fact-sensitive applications and for designing lightweight hallucination mitigation strategies.

Part III: Tipping strategy and RAG ablation experiment

1. Experiment Overview

1.1 Objective

This experiment, through a systematic ablation study, aims to evaluate the effectiveness of different prompting strategies and retrieval enhancement generation techniques in improving the accuracy of the Qwen2.5-7B-Instruct model in question-answering tasks.

1.2 Dataset

We used a subset of the official HaluEval QA dataset, designed specifically for evaluating large language model illusions. Each sample contains fields such as question, knowledge, and right_answer.In the experiments, 50 samples were randomly selected from the entire dataset for testing. The knowledge base uses all knowledge points to build a global RAG index to ensure comprehensive retrieval context.

1.4 Global Generation Parameters

Sample Size: NUM_SAMPLES = 50 (balancing experimental efficiency and statistical significance); **RAG Top-K**: 2 (balancing context length and relevance); **Generation Parameters**: max_new_tokens=256, temperature=0.3, top_p=0.85 (low temperature ensures stable output and reduces random illusions); **Model Loading**: 4-bit NF4 quantization + double quantization + bfloat16 + device_map="auto" (achieving efficient inference on consumer-grade hardware); **Embedded Model**: all-MiniLM-L6-v2 + FAISS IndexFlatL2 (efficient local vector retrieval).

2. Experimental Pipeline

1. Clone and load the HaluEval dataset, randomly selecting 50 samples.
2. Construct a global knowledge base vector using SentenceTransformer and create a FAISS index.
3. Load the 4-bit quantized Qwen2.5-7B-Instruct model.Define 6 experimental configurations and corresponding prompt templates.

4. For each configuration and each sample, perform the following sequentially:(Optional) RAG retrieval and concatenation of numbered context;Construct system + user prompt;Invoke the model to generate answers;Evaluate correctness using simple string matching.Generate a comparison table, a scale-illusion rate curve.
5. Summarize the results, generate a DataFrame, calculate the accuracy for each configuration, and save the complete results to halueval_ablation_results.csv.

3. Results and Analysis

3.1 Overall Performance

Overall, the three configurations incorporating RAG significantly outperformed the group without RAG in accuracy, indicating that external knowledge retrieval can effectively reduce model illusion. The Self-Reflection prompt strategy further enhances the reliability of the answers through a "think-generate-reflect" process, especially when combined with RAG.

3.2 Quantitative Results

Table 2-Accuracy Results Chart

config	accuracy	count
Self-Reflection + RAG	0.80	50
Citation + RAG	0.76	50
Standard + RAG	0.74	50
Citation	0.46	50
Self-Reflection	0.40	50
Standard	0.40	50

4. Key Findings

1. The optimal combination, "Self-Reflection + RAG," achieved the highest accuracy (80.0%), demonstrating that the combination of reflection and external knowledge retrieval is currently the best practice.And RAG technology is the single strongest factor in improving accuracy; all three RAG configurations outperformed their corresponding non-RAG groups.
2. The citation mechanism not only facilitates manual verification but also indirectly improves the model's fidelity to the context.

5. Conclusion of Part III

This ablation experiment system verifies the synergistic benefit of cue engineering and RAG technology in the Qwen2.5-7B-Instruct QA task. The Self-Reflection + RAG configuration exhibits optimal performance under the current settings.

Conclusion & Application strategy

By varying the model parameter size, whether it's open-source or closed-source, and whether or not to add RAG technology, we examined measurements such as illusion rate and accuracy on natural language tasks. Our findings show that, after balancing response length and inference speed, appropriately increasing the number of model parameters, using more efficient training methods and more advanced model architectures, and adding RAG technology can effectively mitigate the model's illusionary behavior. This research is expected to provide practical strategies for building more reliable LLM applications.